

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 1.1: Real Numbers — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.1: Real Numbers
Driving Question	DRIVING QUESTION: How do we use real numbers in everyday life, and why are reciprocals important?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Shopping at the duka: what kinds of numbers?	What we observed in Achieng's trip to the duka: she bought 3 items at Ksh 45 each (total Ksh 135), paid Ksh 200 and got Ksh 65 change; 6 mandazi cost Ksh 90 so the unit price is Ksh 15; 1/2 kg of sugar cost Ksh 80; a tea-leaf sachet tag was Ksh 22.50. Whole numbers, a fraction and a decimal all appeared.	What we learned: numbers come in different types, each with its own purpose: whole numbers for counting items and money, fractions for parts of a whole, and decimals for prices with cents. Fractions and decimals can name the same quantity ($1/2 = 0.5$; $22.50 = 22 \frac{1}{2}$). Unit price is found by dividing the total cost by the number of items.	Our explanation so far: Achieng's trip needs different kinds of numbers because counting, sharing a whole into parts and pricing with cents are different jobs. Unit price is a division, $90 \div 6 = \text{Ksh } 15$. We still wonder how whole numbers can be classified, what other kinds of numbers exist, and why reciprocals matter. These are on our DQB.
Lesson 2 Classifying Whole Numbers: Odd, Even, Prime and Composite	We listed the factors of the duka numbers 3, 6, 15, 45, 65, 90, 135 and 200. We tried to share 17 cobs of maize equally and always had a remainder. We saw that 14 seats (2×7) and 30 eggs ($2 \times 3 \times 5$) can be arranged in several equal ways. We built the sieve of Eratosthenes for 1 to 50 and found 15 primes.	Even numbers are divisible by 2 and odd numbers are not. A prime has exactly two factors (1 and itself) and a composite has more than two factors. 1 is neither prime nor composite, and 2 is the only even prime. Being odd does not make a number prime (15, 45, 65 and 135 are odd composites).	Achieng's duka numbers are classified by counting factors: 3 is prime and odd; 6, 90 and 200 are even composites; 15, 45, 65 and 135 are odd composites. This answers DQB question 3 and adds a classification branch to our unit model of number types, showing that the whole numbers at the duka are not all the same kind.
Lesson 3 Rational and Irrational Numbers: Why Do Some Decimals Never End?	We divided the numerator by the denominator to convert $1/2$, $1/4$, $1/8$, $1/3$, $1/6$, $1/7$, $1/15$ and $1/45$ into decimals. $1/2 = 0.5$, $1/4 = 0.25$, $1/8 = 0.125$ ended; $1/3 = 0.333...$, $1/6 = 0.1666...$, $1/7 = 0.142857142857...$, $1/15 = 0.0666...$, $1/45 = 0.02222...$ repeated. $\sqrt{2} = 1.41421356...$ and $\pi = 3.14159265...$ showed no end and no	Rational numbers can be written as a/b ($b \neq 0$) and have terminating or recurring decimals; whole numbers are rational ($n = n/1$) and $\text{Ksh } 22.50 = 45/2$. Irrational numbers such as $\sqrt{2}$ and π have decimals that never end and never repeat. Rational and irrational numbers together make up the real numbers. Fractions and decimals	One shopping trip at the duka needs whole numbers, fractions and decimals because they are different ways of writing rational numbers for counting, sharing parts and pricing. They are all real numbers. Some numbers like $\sqrt{2}$ and π are irrational because their decimals never end or repeat, and we still need to find out why $\sqrt{2}$ goes on

	repeating block.	are two ways of writing the same rational numbers.	forever (DQB 5).
Lesson 4 Why does $\sqrt{2}$ go on forever? Squaring checks on a square field	We squared 1.414, 1.4142 and 1.415 and got 1.999396, 1.99996164 and 2.002225. The first two are just under 2 and the last is over 2. The side of the square plot of area 2 m ² is between 1.4142 and 1.415. $\sqrt{2} = 1.41421356...$ and the digits do not end or repeat.	No decimal that ends can square to exactly 2, and $\sqrt{2}$ has no repeating block, so $\sqrt{2}$ cannot be written as a fraction a/b : it is an irrational number. Rational numbers (whole numbers, fractions, terminating and recurring decimals) and irrational numbers (such as $\sqrt{2}$ and π) together make up the real numbers.	$\sqrt{2}$ goes on forever because it is irrational: its decimal never ends and never repeats, unlike $1/2 = 0.5$ or $1/7 = 0.142857...$, which are rational. All the numbers on Achieng's duka trip (3, 15, $1/2$, Ksh 22.50) are rational real numbers, and the real-number map now also has a place for irrational numbers such as $\sqrt{2}$.
Lesson 5 What is a Reciprocal? Sharing, Unit Price and Division	We observed that dividing 1 by 2, 4, 8, 6, 15, 45, $1/2$ and 2.5 gives 0.5, 0.25, 0.125, 0.1667, 0.0667, 0.02222, 2 and 0.4, and that each number multiplied by its answer gives 1 (for example $6 \times 1/6 = 1$ and $45 \times 1/45 = 1$). Also $90 \div 6 = 15 = 90 \times 1/6$.	The reciprocal of a number n is $1 \div n$, written $1/n$. It is found by division, and $n \times 1/n = 1$. Zero has no reciprocal. Recurring reciprocals such as $1/6$ are rounded to 4 d.p. or 4 s.f.	Unit price, such as Ksh 15 for one mandazi, is found by dividing the total by the number of items, which is the same as multiplying by the reciprocal ($90 \times 1/6 = 15$). Sharing 1 whole into n equal parts gives $1/n$, and n of those parts make 1 again, so reciprocals turn division problems into multiplication.
Lesson 6 Reciprocals with Mathematical Tables and Calculators: How Accurate Is Each Tool?	We found $1/7$, $1/6$, $1/45$ and $1/2.5$ using division, mathematical tables and the calculator $1/x$ key. Tables gave $1/7 \approx 0.1429$, $1/6 \approx 0.1667$ and $1/45 \approx 0.02222$. The calculator gave $1/7 = 0.142857142...$ and $1/2.5 = 0.4$.	Reciprocals can be found by division, from mathematical tables (4 significant figures) and with the calculator $1/x$ key (more digits). Recurring decimals like $1/7$ must be rounded, while $1/2.5 = 0.4$ ends exactly. We check with $n \times 1/n \approx 1$, and we look after the tables and calculators.	Mzee Odhiambo can quickly find the unit price of the mandazi, $90 \times 1/6 = \text{Ksh } 15$, because the reciprocal of 6 is 0.1667 (rounded), and the tools give results that are accurate enough for money. The tables give fewer digits than the calculator because $1/6$ and $1/7$ are recurring decimals that never end, so every tool has to round them.
Lesson 7 Reciprocals in Speed, Time, Rate and Interest: Answering the Driving Question	We observed that a journey of 480 km takes 8 hours at 60 km/h and 6 hours at 80 km/h ($480 \times 1/60 = 8$, $480 \times 1/80 = 6$), that Wanjiru's Ksh 10 000 at 8% earns Ksh 800 a year and Ksh 400 in 6 months, and that Ksh 2 400 interest takes $2\,400 \times 1/800 = 3$ years. We also saw that the duka unit price $90 / 6 = \text{Ksh } 15$ is the same as $90 \times 1/6$.	We learned that real numbers include whole numbers, fractions and decimals (rational numbers) and numbers like $\sqrt{2}$ and π (irrational numbers); that whole numbers are odd, even, prime or composite; that the reciprocal of n is $1/n$ and $n \times 1/n = 1$; and that reciprocals can be found by division, mathematical tables or the calculator $1/x$ key.	We explain that Achieng's one shopping trip needs whole numbers for counting, fractions for parts and decimals for prices because they are different ways of writing real numbers, and that reciprocals are important because they turn division into multiplication in unit price, speed, time, rate and interest problems, so real numbers and reciprocals together describe everyday life.

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